ELECTRICAL & AUTOMATION PORTFOLIO

Low-Voltage Distribution & PLC Control Panel Assembly

A selection of switchgear and control panels designed, wired, and commissioned during the Melemco building-automation internship — covering a 400A distribution panel, ATS control, power-factor correction, and PLC-based panels.

400A Distribution Panel · PLC Panels · ATS Panel · MCCB & Capacitor Bank · Fuses & Segregated Breakers

CONTENTS

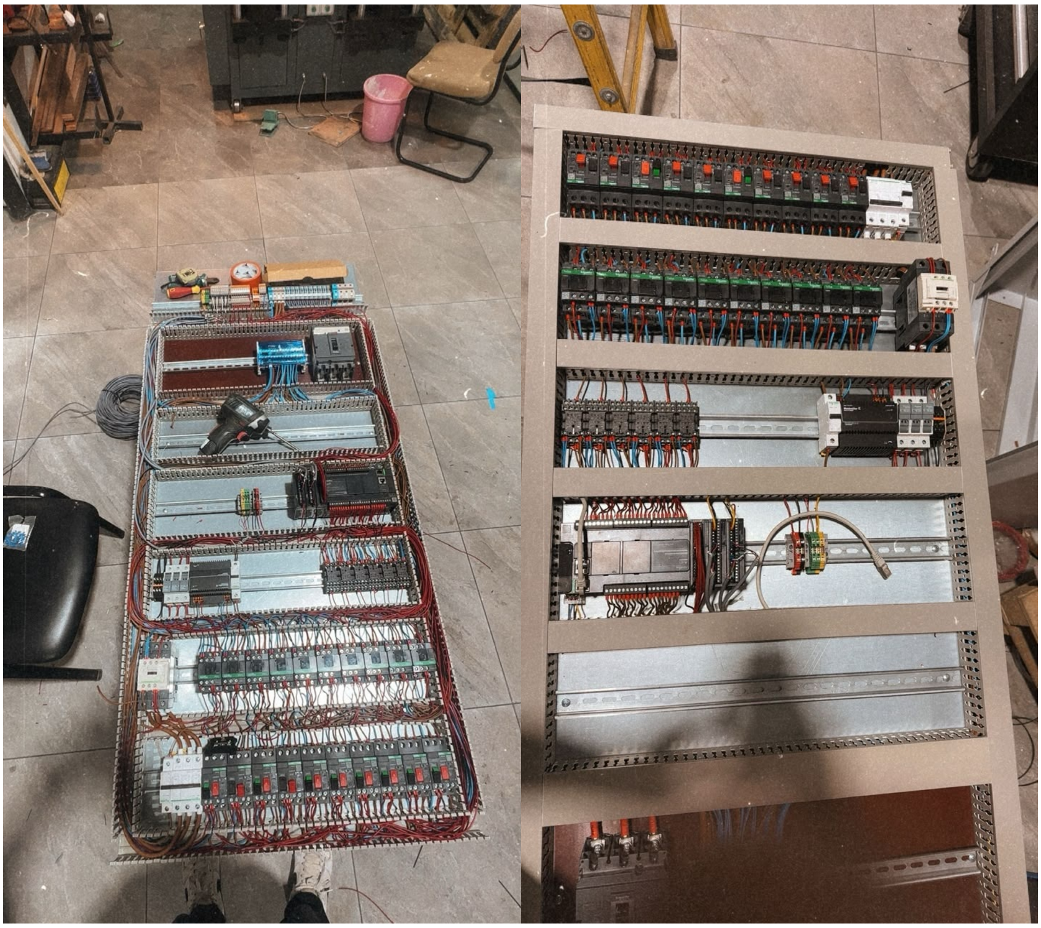
PREPARED BY
Karim

COMPANY
Melemco Modern Electromechanical Eng. Co.

SCOPE
Panel Build & Wiring

Fig. 1 — PLC Panel: Full Assembly & Relay/Breaker Bank

PLC control panel shown at two build stages — the complete bench assembly, and the CPU/I-O section with interposing relays and dedicated breaker banks for field devices.

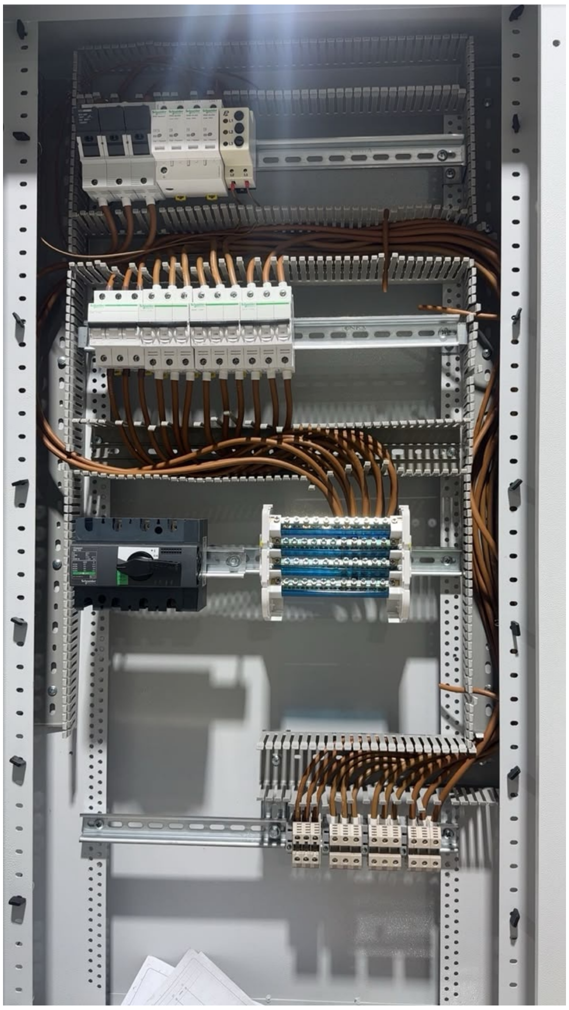


KEY ELEMENTS

- PLC CPU and I/O modules mounted alongside the distribution devices
- Interposing relay banks isolating field signals from control logic
- Terminal strips and wiring ducts organizing incoming and outgoing circuits
- Dedicated MCB rows protecting individual output circuits
- Neatly dressed and numbered control wiring across all DIN rails

Fig. 2 — Fuses & Segregated Breaker Section

Interior view of the enclosure showing surge protection, fuse-protected feeders, and a segregated breaker section feeding the outgoing terminal blocks.

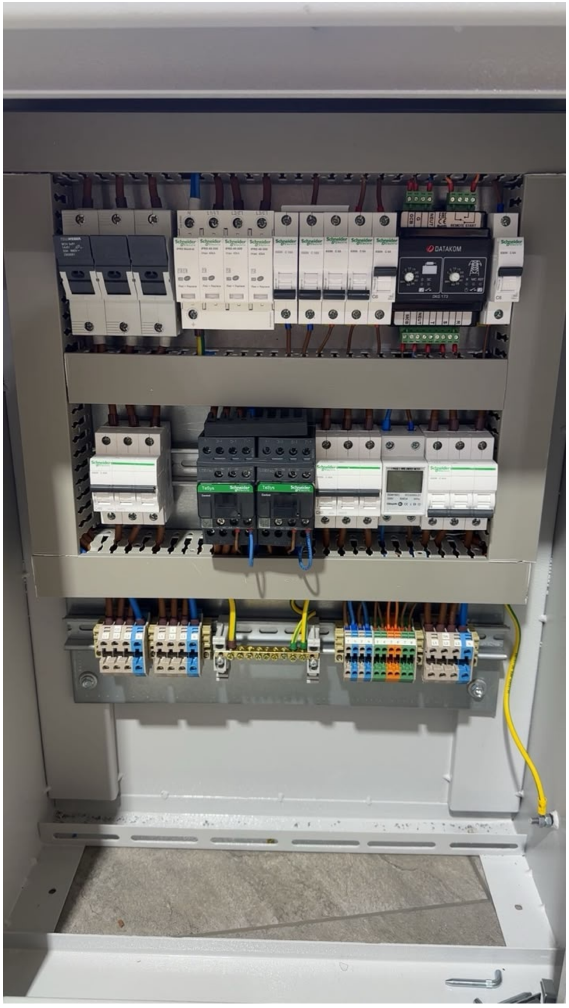


KEY ELEMENTS

- Type 2 surge protection device (SPD) with remote-signaling contact
- Fuses protecting individual outgoing feeders
- Segregated breaker bank keeping circuits physically and electrically separated
- Consistent brown-conductor wiring routed through slotted trunking

Fig. 3 — Automatic Transfer Switch (ATS) Panel

ATS control section combining metering, a generator/mains protection relay, and switching contactors for automatic source transfer.

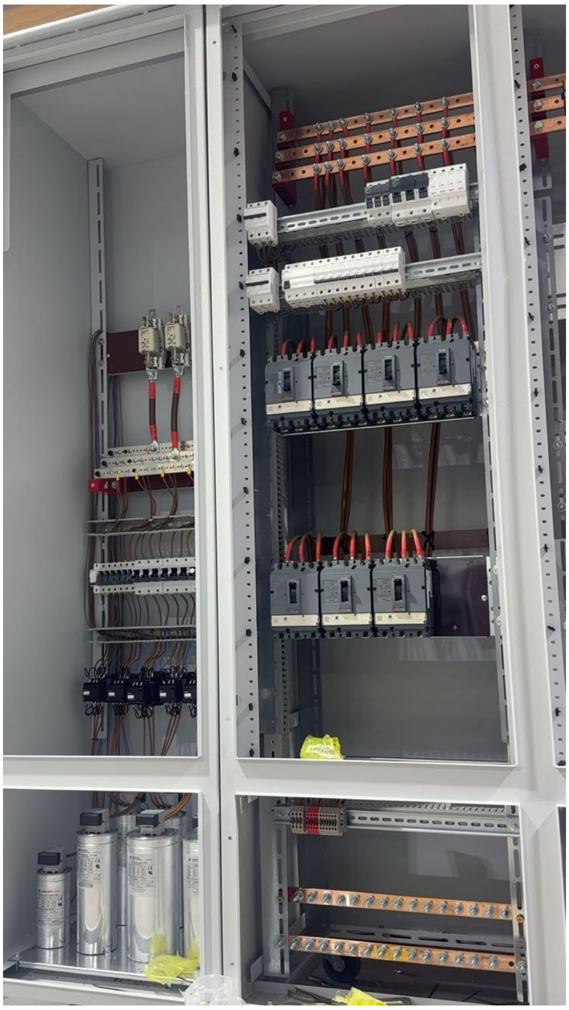


KEY ELEMENTS

- Datakom protection relay for generator/mains monitoring and remote start
- Schneider TeSys contactors for automatic mains/generator transfer
- Digital energy meter with branch MCB protection
- Color-coded control terminals (blue/orange/black) for clear signal routing

Fig. 4 — 400A Panel: MCCBs & Capacitor Bank

400A distribution cabinet housing moulded-case circuit breakers (MCCBs) for main feeders and the power-factor-correction capacitor bank.



KEY ELEMENTS

- Copper busbar system rated for 400A distributing power to multiple MCCB feeders
- Fuse-based isolators for outgoing circuit protection
- Automatic power-factor-correction (APFC) capacitor bank
- Structured cable management maintaining clear bend radii on heavy feeders

Skills Demonstrated

Practical panel-building and automation skills gained through hands-on assembly of these enclosures.

Panel Wiring & Layout

DIN-rail component layout, wire duct routing, conductor color-coding, and terminal numbering for maintainable, standards-aligned panels.

Protection & Switchgear

Selection and wiring of MCBs, MCCBs, fuse-switches, and surge protection devices for low-voltage distribution.

PLC Integration

Mounting and field-wiring of PLC CPU/I-O modules, interposing relays, and network connections within control panels.

Power Quality

Installation of automatic power-factor-correction capacitor banks and protection relays for generator/mains switching.

ATS & Transfer Control

Contactor-based automatic transfer switch wiring using Schneider TeSys components with associated protection relays.

Metering & Monitoring

Integration of digital energy meters and Datakom relays for real-time monitoring of panel loads and supply status.